

CSS Box Model & box-sizing: border-box

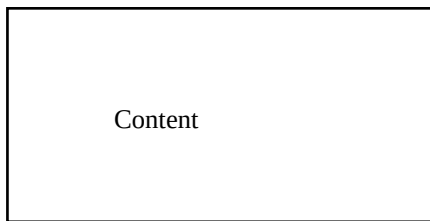
The **box-sizing** property in CSS controls how the total width and height of an element are calculated. Understanding this is very important for layout design.

1. Default box-sizing: content-box

- This is the default behavior in CSS.
- Width and height apply **only to content**.
- Padding and border are added **outside** the defined width/height.

Result:

When padding is added, the **overall size increases**, but the content size stays the same.



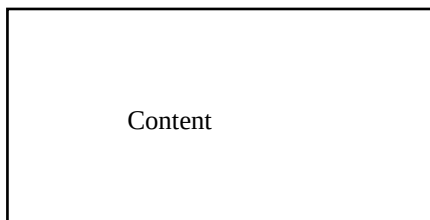
Padding + Border increase size

2. box-sizing: border-box

- Width and height include **content + padding + border**.
- The overall size remains **fixed**.
- Padding is adjusted by shrinking the content area.

Result:

When padding is added, the **content shrinks**, but the border size remains the same.



Content shrinks, size fixed

3. Quick Comparison

Property	content-box (default)	border-box
Content Size	Fixed	Shrinks when padding added

Padding Effect	Increases total size	Included inside size
Border Size	Increases total size	Remains same
Layout Control	Harder	Easier & predictable

4. Example Code

```
div {  
  width: 200px;  
  padding: 20px;  
  box-sizing: border-box;  
}
```

Conclusion: Using **box-sizing: border-box** is recommended for modern layouts because it keeps element sizes predictable and prevents layout breaking.